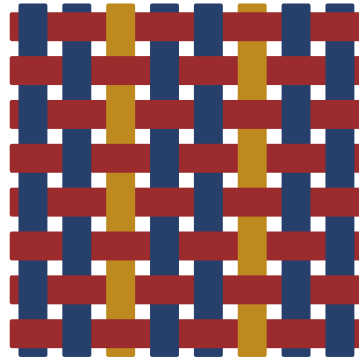




Information Theory

Lecture 1 · from surprise to error bounds

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THE ONE IDEA, AND THE MAP

□□□□

A probability law tells us which outcomes are likely. Information theory turns this into **log-arithmetic surprise**, then averages, compares, and conditions it to obtain entropy, divergence, and mutual information. The rest of the lecture establishes what these quantities can do under comparison and processing, and ends by turning residual uncertainty into a lower bound on estimation error.

The map: eight stops from surprise to error bounds

stop	idea	what it unlocks
1	$-\log p(x)$: surprise	turns a probability law into an additive information scale
2	$H(X)$: average surprise	quantifies uncertainty and identifies the maximum-entropy law
3	$D(p\ q)$: model mismatch	compares a true law with a reference law, in a directed way
4	$I(X; Y)$ and $H(Y X)$	separate shared information from uncertainty that remains
5	log-sum and divergence inequalities	give non-negativity, chain rules, and convexity
6	conditional mutual information	yields data processing: transformations cannot create information
7	entropy under constraints	gives range, equality cases, and the effect of deterministic maps or mixing
8	Fano's inequality	converts residual uncertainty into a lower bound on decoding error

Notation used throughout

Unless a base is shown, $\log = \log_2$ and information is measured in bits. For zero-probability terms we use $0 \log 0 = 0$.

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LECTURE 1 · FROM SURPRISE TO ERROR BOUNDS

Build the four basic quantities from a probability law: surprise, entropy, divergence, and mutual information, with conditional entropy as the bridge. Then use log-sum, non-negativity, and chain rules to show that processing cannot create information, before turning decoding errors into entropy bounds.

■ 1 PROBABILITY LAWS AND SURPRISE

□□□□ The probability law is the object; surprise is a logarithm of that law.

Before measuring information, keep the random variable separate from its law. For a discrete random variable, the probability mass function records the chance of every possible value. The same law may also be viewed as a probability measure or through its cumulative distribution function.

Three views of the law

- **Definition 1.1** (*probability mass function*). Let X take values in a discrete alphabet $\mathcal{X}$. Its probability mass function is

$$p : \mathcal{X} \longrightarrow [0, 1], \quad p(x) = \mathbb{P}(X = x).$$

We also write p_X when the random variable must be made explicit.

- **Definition 1.2** (*distribution and CDF*). The distribution of a random variable X is denoted P_X . For any set A in the value space, define

$$P_X(A) = \mathbb{P}(X \in A).$$

If X is a real-valued random variable, its cumulative distribution function is defined as

$$F_X(y) = \mathbb{P}(X \leq y), \quad y \in \mathbb{R}.$$

Your turn. Complete the dictionary:

$$p_X(x) = \underline{\mathbb{P}(X = x)}, \quad P_X((a, b)) = \underline{\mathbb{P}(a < X < b)}.$$

Why surprise is logarithmic

?

$p_X(x) = \mathbb{P}(X = x)$
and $P_X((a, b)) = \mathbb{P}(a < X < b)$.



Definition 1.3 (*surprise of an event*). We seek a function $s(A)$ that depends continuously on $P(A)$, decreases as $P(A)$ increases, and satisfies

$$s(A \cap B) = s(A) + s(B)$$

whenever A and B are independent. Up to the choice of logarithm base, these requirements give

$$s(A) = -\log P(A).$$

Your turn. If independent events have probabilities $P(A) = 1/2$ and $P(B) = 1/4$, then

$$s(A) = \underline{1}, \quad s(B) = \underline{2}, \quad s(A \cap B) = \underline{3} \text{ bits.}$$

? 1, 2, and 3 bits. Independence gives $P(A \cap B) = P(A)P(B) = 1/8$.

2 ENTROPY: AVERAGE SURPRISE

□□□□ Entropy averages the surprise of all possible outcomes.

The surprise $-\log p(x)$ belongs to one outcome. Entropy takes its expectation, so it depends on the *law* of X , not on which value happens to be observed.



Definition 2.1 (*entropy in base b*). For a discrete random variable X with alphabet $\mathcal{X}$,

$$H_b(X) = - \sum_{x \in \mathcal{X}} p_X(x) \log_b p_X(x) = \mathbb{E}[-\log_b p_X(X)].$$

We use the convention $0 \log_b 0 = 0$. Unless stated otherwise, $b = 2$.

Remark 2.2. Because entropy depends only on p_X , the notations $H(X)$, $H(p_X)$, and $H(p)$ may describe the same number.

A Bernoulli random variable



Example 2.3 (*binary entropy*). Let $\mathcal{X} = \{H, T\}$ with $P(X = H) = p$ and $P(X = T) = 1 - p$. Then

$$H(X) = -p \log p - (1 - p) \log(1 - p).$$

In this special case we sometimes write $H(p)$.

Your turn. Use the formula to fill the three anchor values:

$$H(0) = \underline{0}, \quad H\left(\frac{1}{2}\right) = \underline{1}, \quad H(1) = \underline{0} \quad \text{bits.}$$

Sketch the curve for $0 \leq p \leq 1$ and mark its highest point.

? The values are 0, 1, 0. The curve is symmetric about $p = 1/2$ and peaks there.

Joint entropy

Definition 2.4 (*joint entropy*). For $X = (X_1, X_2)$ with joint pmf $p_{X_1 X_2}$,

$$H(X_1, X_2) = - \sum_{x_1 \in \mathcal{X}_1} \sum_{x_2 \in \mathcal{X}_2} p_{X_1 X_2}(x_1, x_2) \log p_{X_1 X_2}(x_1, x_2).$$

Proposition 2.5 (*additivity under independence*). If X_1 and X_2 are independent, then

$$H(X_1, X_2) = H(X_1) + H(X_2).$$

Proof. Independence gives $p_{X_1 X_2}(x_1, x_2) = p_{X_1}(x_1)p_{X_2}(x_2)$. Insert this into Definition 2.4 and split the logarithm:

$$\begin{aligned} H(X_1, X_2) &= - \sum_{x_1, x_2} p_{X_1}(x_1)p_{X_2}(x_2)(\log p_{X_1}(x_1) + \log p_{X_2}(x_2)) \\ &= - \sum_{x_1} p_{X_1}(x_1) \log p_{X_1}(x_1) \sum_{x_2} p_{X_2}(x_2) \\ &\quad - \sum_{x_2} p_{X_2}(x_2) \log p_{X_2}(x_2) \sum_{x_1} p_{X_1}(x_1) \\ &= H(X_1) + H(X_2), \end{aligned}$$

because each marginal pmf sums to 1. ■

Your turn. Recall the key step in the additivity proof:

$$\log(p_{X_1}(x_1)p_{X_2}(x_2)) = \underline{\log p_{X_1}(x_1) + \log p_{X_2}(x_2)}.$$

? $\log p_{X_1}(x_1) + \log p_{X_2}(x_2)$.

3 DIVERGENCE: COMPARING TWO LAWS

□□□□ KL divergence is not a distance, but distance intuition is useful.

$D(p\|q)$ measures the inefficiency that arises when the true law is p but we assume the law is q . It is asymmetric, so it is not a distance between distributions in the strict mathematical sense. Nevertheless, treating it as a directed notion of "distance" is often very useful.

Definition 3.1 (Kullback–Leibler divergence). For pmfs p and q on $\mathcal{X}$,

$$D(p\|q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)}.$$

We take $0 \log(0/q) = 0$. If $p(x) > 0$ for some x with $q(x) = 0$, then $D(p\|q) = \infty$.

Expectation and entropy forms

If $X \sim p$, then

$$\begin{aligned} D(p\|q) &= \mathbb{E}_{X \sim p} \left[\log \frac{p(X)}{q(X)} \right] \\ &= -\mathbb{E}_{X \sim p} [\log q(X)] - H(X). \end{aligned}$$

Your turn. Fill the algebraic step:

$$\mathbb{E}_p \left[\log \frac{p(X)}{q(X)} \right] = \mathbb{E}_p [\log p(X)] - \mathbb{E}_p [\log q(X)] = \underline{-H(X) - \mathbb{E}_p [\log q(X)]}.$$

? $-H(X) - \mathbb{E}_p [\log q(X)]$.

The order cannot be reversed

Example 3.2 (asymmetry). On $\mathcal{X} = \{0, 1\}$, let

$$p = (1/2, 1/2), \quad q = (1, 0).$$

Then $D(p\|q) = \infty$ because $q(1) = 0$ while $p(1) > 0$, but

$$D(q\|p) = \log \frac{1}{1/2} = 1 \text{ bit}.$$

Thus $D(p\|q)$ and $D(q\|p)$ need not agree.

Your turn. Explain in one sentence why the first direction is infinite but the reverse direction is finite.

$D(p\|q)$ sums $p(x) \log \frac{p(x)}{q(x)}$; since $p(1) > 0$ while $q(1) = 0$, the term $\log \frac{p(1)}{q(1)} = \log \infty$ blows up. In the reverse direction every x with $q(x) > 0$ also has $p(x) > 0$, so no infinite term appears.

? $p(1) > 0 = q(1)$ makes $D(p\|q) = \infty$; the reversed support has no such term.

Comparison with the uniform law

Proposition 3.3 (uniform reference). If q is uniform on a finite alphabet $\mathcal{X}$, then for $X \sim p$,

$$D(p\|q) = \log |\mathcal{X}| - H(X).$$

Proof. Since $q(x) = 1/|\mathcal{X}|$,

$$\begin{aligned} D(p\|q) &= \sum_x p(x) \log p(x) + \sum_x p(x) \log |\mathcal{X}| \\ &= -H(X) + \log |\mathcal{X}| \sum_x p(x) \\ &= \log |\mathcal{X}| - H(X), \end{aligned}$$

where the last equality uses $\sum_x p(x) = 1$. ■

Your turn. Recall the key simplification for a uniform reference law:

$$D(p\|q) = -H(X) + \log |\mathcal{X}| \sum_x p(x) = \underline{\log |\mathcal{X}| - H(X)}.$$

? $\log |\mathcal{X}| - H(X)$.

4 SHARED AND REMAINING UNCERTAINTY

□□□□ Mutual information measures how much one random variable tells us about another; conditional entropy measures what remains.

Mutual information $I(X; Y)$ is a measure of the amount of information one random variable contains about another; equivalently, it is the reduction in uncertainty about one random variable once the other is known.

Definition 4.1 (mutual information). For discrete X and Y taking values in $\mathcal{X}$ and $\mathcal{Y}$,

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{XY}(x, y) \log \frac{p_{XY}(x, y)}{p_X(x)p_Y(y)}.$$

Equivalently,

$$I(X; Y) = D(p_{XY} \| p_X p_Y).$$

Entropy form

Taking expectation with respect to the joint law,

$$\begin{aligned}
I(X; Y) &= \mathbb{E} \left[\log \frac{p_{XY}(X, Y)}{p_X(X)p_Y(Y)} \right] \\
&= H(X) + H(Y) - H(X, Y).
\end{aligned}$$

Indeed, the three expectations in the log-ratio are

$$\mathbb{E}[\log p_{XY}(X, Y)] - \mathbb{E}[\log p_X(X)] - \mathbb{E}[\log p_Y(Y)] = -H(X, Y) + H(X) + H(Y).$$

Conditional entropy



Definition 4.2 (*conditional entropy*). The conditional entropy of Y given X is

$$H(Y | X) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{XY}(x, y) \log p_{Y|X}(y | x).$$

Equivalently, it is the average of the entropies of the conditional laws:

$$H(Y | X) = \sum_{x \in \mathcal{X}} p_X(x) H(Y | X = x).$$



Proposition 4.3 (*chain rule*).

$$H(Y | X) = H(X, Y) - H(X), \quad H(X | Y) = H(X, Y) - H(Y).$$

Proof. Bayes' rule gives $p_{Y|X}(y | x) = p_{XY}(x, y)/p_X(x)$. Substituting this ratio into Definition 4.2 gives

$$\begin{aligned}
H(Y | X) &= - \sum_{x,y} p_{XY}(x, y) \log \frac{p_{XY}(x, y)}{p_X(x)} \\
&= - \sum_{x,y} p_{XY}(x, y) \log p_{XY}(x, y) + \sum_{x,y} p_{XY}(x, y) \log p_X(x) \\
&= H(X, Y) + \sum_x p_X(x) \log p_X(x) \\
&= H(X, Y) - H(X).
\end{aligned}$$

Interchanging X and Y yields $H(X | Y) = H(X, Y) - H(Y)$. ■

Your turn. Recall the key logarithm split in the chain-rule proof:

$$\log(p_X(x)p_{Y|X}(y | x)) = \underline{\log p_X(x) + \log p_{Y|X}(y | x)}.$$

? $\log p_X(x) + \log p_{Y|X}(y | x)$.

All identities on one line

The definitions now meet:

$$I(X; Y) = H(X) + H(Y) - H(X, Y) = H(Y) - H(Y | X) = H(X) - H(X | Y).$$

■ 5 DIVERGENCE INEQUALITIES

□□□□ The log-sum inequality turns local ratios into global comparisons.

KL divergence is not merely a formula. Its most useful properties come from one move: group positive terms, apply convexity to their ratios, and keep track of the equality condition. This is the engine behind non-negativity, chain rules, coarse-graining, and convexity.

■ **Lemma 5.1** (*information inequality*). For pmfs p and q on the same alphabet,

$$D(p\|q) \geq 0,$$

with equality if and only if $p = q$.

Proof. If $q(x) = 0$ for some x with $p(x) > 0$, then $D(p\|q) = +\infty$. Otherwise, let $X \sim p$ and $S = \{x : p(x) > 0\}$. Jensen's inequality for the convex function $-\log$ gives

$$\begin{aligned} D(p\|q) &= \mathbb{E}_p \left[-\log \frac{q(X)}{p(X)} \right] \\ &\geq -\log \mathbb{E}_p \left[\frac{q(X)}{p(X)} \right] = -\log \sum_{x \in S} q(x) \geq -\log 1 = 0. \end{aligned}$$

Equality in Jensen requires $q(x)/p(x)$ to be constant on S , while equality in the second inequality requires q to put no mass outside S . Normalization then forces that constant to be 1, so $p = q$. ■

■ **Theorem 5.2** (*log-sum inequality*). For non-negative numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$, with $a = \sum_i a_i$ and $b = \sum_i b_i$,

$$\sum_{i=1}^n a_i \log \frac{a_i}{b_i} \geq a \log \frac{a}{b}.$$

If all relevant terms are positive, equality holds exactly when a_i/b_i is constant in i .

Proof. The boundary cases follow from the usual conventions for zero terms. In the non-trivial finite case $a, b > 0$, put $\alpha_i = a_i/a$ and $\beta_i = b_i/b$. Then

$$\begin{aligned} \sum_i a_i \log \frac{a_i}{b_i} &= a \log \frac{a}{b} + a \sum_i \alpha_i \log \frac{\alpha_i}{\beta_i} \\ &= a \log \frac{a}{b} + a D(\alpha\|\beta) \\ &\geq a \log \frac{a}{b}, \end{aligned}$$

where the last step is Lemma 5.1. Equality holds precisely when $\alpha = \beta$, equivalently $a_i/b_i = a/b$ on the common support. ■

Your turn. What condition makes log-sum an equality?

$$\frac{a_1}{b_1} = \dots = \frac{a_n}{b_n} = \underline{\frac{a}{b}}.$$

? Equality holds when every ratio is the common value a/b .

A chain rule for model mismatch

For joint pmfs $p_{XY} = p_X p_{Y|X}$ and $q_{XY} = q_X q_{Y|X}$, define the conditional divergence

$$D(p_{Y|X} \| q_{Y|X} \mid p_X) := \sum_x p_X(x) D(p_{Y|X}(\cdot \mid x) \| q_{Y|X}(\cdot \mid x)).$$

Proposition 5.3 (chain rule for divergence).

$$D(p_{XY} \| q_{XY}) = D(p_X \| q_X) + D(p_{Y|X} \| q_{Y|X} \mid p_X).$$

Proof. Factor both joint laws and split the logarithm:

$$\begin{aligned} D(p_{XY} \| q_{XY}) &= \sum_{x,y} p_{XY}(x,y) \log \frac{p_X(x) p_{Y|X}(y \mid x)}{q_X(x) q_{Y|X}(y \mid x)} \\ &= \sum_x p_X(x) \log \frac{p_X(x)}{q_X(x)} + \sum_x p_X(x) \sum_y p_{Y|X}(y \mid x) \log \frac{p_{Y|X}(y \mid x)}{q_{Y|X}(y \mid x)}. \end{aligned}$$

These are precisely the two terms in the claim. ■

Corollary 5.4 (marginalization cannot increase divergence).

$$D(p_{XY} \| q_{XY}) \geq D(p_X \| q_X).$$

Proof. The difference is the conditional divergence in Proposition 5.3, which is an average of non-negative divergences. ■

Mixing pairs of distributions

Proposition 5.5 (joint convexity). For pmfs p_1, p_2, q_1, q_2 and $0 \leq \lambda \leq 1$,

$$\begin{aligned} D(\lambda p_1 + (1 - \lambda) p_2 \| \lambda q_1 + (1 - \lambda) q_2) \\ \leq \lambda D(p_1 \| q_1) + (1 - \lambda) D(p_2 \| q_2). \end{aligned}$$

Proof. For each outcome x , apply log-sum to $a_1 = \lambda p_1(x)$, $a_2 = (1 - \lambda) p_2(x)$ and the corresponding b_1, b_2 built from q_1, q_2 . Summing the resulting inequality over x gives the claim. ■

Your turn. In the divergence chain rule, what remains after subtracting the marginal term?

$$D(p_{XY} \| q_{XY}) - D(p_X \| q_X) = \underline{D(p_{Y|X} \| q_{Y|X} \mid p_X)}.$$

? The remainder is an average conditional divergence, so it is non-negative.

6 INFORMATION CAN ONLY BE LOST

□□□□ Condition, expand, and discard a non-negative term.

Mutual information behaves like shared evidence. It is non-negative and symmetric, it can be revealed one variable at a time, and processing an observation cannot manufacture information about its source.

The core identities

■ **Proposition 6.1** (*information properties*). For discrete random variables X and Y ,

$$I(X; Y) \geq 0.$$

Equality holds if and only if X and Y are independent. Moreover,

$$I(X; Y) = I(Y; X),$$

$$I(X; Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X).$$

Proof. By Definition 4.1,

$$I(X; Y) = D(p_{XY} \| p_X p_Y).$$

The information inequality gives non-negativity, with equality exactly when $p_{XY} = p_X p_Y$, which is independence. Symmetry follows because the defining log-ratio is unchanged when X and Y are exchanged. The entropy identities were obtained from the chain rule in Proposition 4.3. ■

■ **Definition 6.2** (*conditional mutual information*). The information shared by X and Y after Z is known is

$$I(X; Y \mid Z) := \sum_z p_Z(z) D(p_{XY|Z}(\cdot, \cdot \mid z) \| p_{X|Z}(\cdot \mid z) p_{Y|Z}(\cdot \mid z)).$$

Hence $I(X; Y \mid Z) \geq 0$.

■ **Proposition 6.3** (*chain rule for mutual information*).

$$I(X_1, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y \mid X_1, \dots, X_{i-1}).$$

Proof. For each term, use the conditional entropy identity

$$I(U; Y | V) = H(Y | V) - H(Y | U, V),$$

with $U = X_i$ and $V = (X_1, \dots, X_{i-1})$. The adjacent conditional entropies telescope:

$$\begin{aligned} \sum_{i=1}^n I(X_i; Y | X_1, \dots, X_{i-1}) &= \sum_{i=1}^n [H(Y | X_1, \dots, X_{i-1}) - H(Y | X_1, \dots, X_i)] \\ &= H(Y) - H(Y | X_1, \dots, X_n). \end{aligned}$$

The final expression is $I(X_1, \dots, X_n; Y)$. ■

Your turn. Expand the two-variable case:

$$I(X_1, X_2; Y) = I(X_1; Y) + \underline{I(X_2; Y | X_1)}.$$

? The missing term is $I(X_2; Y | X_1)$.

Data processing

Definition 6.4 (Markov chain). We write $X \rightarrow Y \rightarrow Z$ when X and Z are conditionally independent given Y , equivalently

$$p_{Z|XY}(z | x, y) = p_{Z|Y}(z | y).$$

Theorem 6.5 (data processing inequality). If $X \rightarrow Y \rightarrow Z$ is a Markov chain, then

$$I(X; Z) \leq I(X; Y).$$

Proof. Expand the same mutual information in two orders:

$$I(X; Y, Z) = I(X; Y) + I(X; Z | Y) = I(X; Z) + I(X; Y | Z).$$

The Markov condition gives $I(X; Z | Y) = 0$, whereas conditional mutual information is non-negative. Therefore

$$I(X; Y) = I(X; Z) + I(X; Y | Z) \geq I(X; Z).$$
■

Corollary 6.6 (deterministic processing). For any function f ,

$$I(X; f(Y)) \leq I(X; Y).$$

Proof. Because $f(Y)$ is determined by Y , the variables form the Markov chain $X \rightarrow Y \rightarrow f(Y)$. Apply Theorem 6.5. ■

Your turn. Name the two proof moves behind data processing:

(1) Expand $I(X; Y, Z)$ in two orders. (2) Use the Markov condition $I(X; Z | Y) = 0$ and non-negativity $I(X; Y | Z) \geq 0$.

? Two moves:
(1) expand $I(X; Y, Z)$ in two orders; (2) use $I(X; Z | Y) = 0$ and $I(X; Y | Z) \geq 0$.

7 ENTROPY UNDER CONSTRAINTS

□□□□ Support, conditioning, and deterministic maps all limit uncertainty.

Entropy cannot vary arbitrarily. A finite alphabet fixes its range, conditioning can only remove uncertainty, and a deterministic map can only merge outcomes. The equality cases tell us exactly when no information was lost.

Range and equality cases

Proposition 7.1 (*entropy bounds*). If X takes values in a finite alphabet $\mathcal{X}$, then

$$0 \leq H(X) \leq \log |\mathcal{X}|.$$

The lower bound is attained exactly when X is constant, and the upper bound is attained exactly when X is uniform on $\mathcal{X}$.

Proof. Each summand $-p_X(x) \log p_X(x)$ is non-negative, which proves the lower bound. Equality requires one outcome to have probability 1.

For the upper bound, let u be uniform on $\mathcal{X}$. The uniform-reference identity from Proposition 3.3 gives

$$D(p_X \| u) = \log |\mathcal{X}| - H(X) \geq 0.$$

Equality in the information inequality holds exactly when $p_X = u$. ■

Your turn. Complete the identity that proves the upper bound:

$$D(p_X \| u) = \log |\mathcal{X}| - \underline{H(X)}.$$

? Use $D(p_X \| u) = \log |\mathcal{X}| - H(X) \geq 0$.

Proposition 7.2 (*conditioning reduces entropy*). For discrete random variables X, Y ,

$$0 \leq H(X | Y) \leq H(X).$$

Moreover,

$$H(X | Y) = H(X) \iff X \text{ and } Y \text{ are independent,}$$

whereas

$$H(X | Y) = 0 \iff X = f(Y) \text{ almost surely}$$

for some function f .

Proof. The lower bound follows because conditional entropy is an average of non-negative entropies:

$$H(X | Y) = \sum_y p_Y(y) H(X | Y = y) \geq 0.$$

It is zero exactly when every conditional law with $p_Y(y) > 0$ is a point mass, which is equivalent to $X = f(Y)$ almost surely.

For the upper bound, Proposition 6.1 gives

$$H(X) - H(X | Y) = I(X; Y) \geq 0.$$

Equality holds exactly when $I(X; Y) = 0$, hence exactly when X and Y are independent. ■

Your turn. Match the two equality cases:

$$\begin{aligned} H(X | Y) = H(X) &\iff \underline{X \perp Y}, \\ H(X | Y) = 0 &\iff \underline{X = f(Y) \text{ a.s.}}. \end{aligned}$$

? The upper equality means independence; zero conditional entropy means X is determined by Y .

■ **Proposition 7.3** (*entropy chain rule and subadditivity*). For discrete $X_1, \dots, X_n$,

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1}) \leq \sum_{i=1}^n H(X_i).$$

Equality in the inequality holds exactly when the variables are mutually independent.

Proof. Repeatedly apply the two-variable chain rule $H(U, V) = H(U) + H(V | U)$. Then use

$$H(X_i | X_1, \dots, X_{i-1}) \leq H(X_i)$$

term by term. Equality means that every X_i is independent of its predecessors, which is equivalent to mutual independence. ■

Deterministic maps and collisions

■ **Proposition 7.4** (*a function cannot increase entropy*). For any function f ,

$$H(f(X)) \leq H(X).$$

Equality holds exactly when f is injective on the support of X .

Proof. Because $f(X)$ is determined by X ,

$$H(X, f(X)) = H(X) + H(f(X) | X) = H(X).$$

Expanding in the other order gives

$$H(X, f(X)) = H(f(X)) + H(X | f(X)) \geq H(f(X)).$$

Equality holds exactly when X can be recovered from $f(X)$, which says that f is injective on the support of X . ■

Your turn. Which non-negative term measures the information lost by f ?

$$H(X) - H(f(X)) = \underline{H(X | f(X))}.$$

? The lost information is $H(X | f(X))$.

■ **Proposition 7.5** (*collision probability*). Let X' be an independent copy of X . Then

$$\mathbb{P}(X = X') \geq 2^{-H(X)}.$$

Equality holds exactly when X is uniform on its support.

Proof. Jensen's inequality for the convex function $t \mapsto 2^t$ gives

$$2^{-H(X)} = 2^{\mathbb{E}[\log p_X(X)]} \leq \mathbb{E}[2^{\log p_X(X)}] = \mathbb{E}[p_X(X)].$$

Independence and identical distribution give

$$\mathbb{E}[p_X(X)] = \sum_x p_X(x)^2 = \mathbb{P}(X = X').$$

Jensen is an equality exactly when $\log p_X(X)$ is constant on the support, that is, when X is uniform there. ■

Your turn. Complete the collision identity:

$$\mathbb{E}[p_X(X)] = \sum_x p_X(x)^2 = \underline{\mathbb{P}(X = X')}.$$

? For an independent copy X' , $\sum_x p_X(x)^2 = \mathbb{P}(X = X')$.

Mixing distributions

■ **Proposition 7.6** (*concavity of entropy*). For pmfs p, q on the same alphabet and $0 \leq \lambda \leq 1$,

$$H(\lambda p + (1 - \lambda)q) \geq \lambda H(p) + (1 - \lambda)H(q).$$

Proof. The function $g(t) = -t \log t$ is concave on $[0, 1]$. Hence, for every x ,

$$g(\lambda p(x) + (1 - \lambda)q(x)) \geq \lambda g(p(x)) + (1 - \lambda)g(q(x)).$$

Summing over x proves the claim. ■

Your turn. Name the one-variable function behind entropy's concavity:

$$g(t) = \underline{-t \log t}.$$

? Entropy is a sum of the concave terms $g(t) = -t \log t$.

■ 8 FANO'S INEQUALITY

□□□□ A decoding error leaves a controlled amount of uncertainty.

Suppose Y is used to guess X . Fano's inequality converts the probability of a wrong guess into an upper bound on the uncertainty that remains about X after Y is observed.

Definition 8.1 (*error indicator*). Assume that X and Y take values in the same finite alphabet $\mathcal{X}$, with $|\mathcal{X}| \geq 2$. Define

$$Z = \mathbf{1}_{\{X \neq Y\}}, \quad P_e = \mathbb{P}(X \neq Y) = \mathbb{P}(Z = 1).$$

The binary entropy function is

$$h_2(t) = -t \log t - (1 - t) \log(1 - t).$$

Theorem 8.2 (*Fano's inequality*).

$$H(X | Y) \leq h_2(P_e) + P_e \log(|\mathcal{X}| - 1).$$

Equivalently,

$$H(X | Y) \leq H(Z) + \mathbb{P}(X \neq Y) \log(|\mathcal{X}| - 1).$$

Proof. The indicator Z is determined by (X, Y) , so $H(Z | X, Y) = 0$. Apply the chain rule in two orders:

$$\begin{aligned} H(X | Y) &= H(X, Z | Y) \\ &= H(Z | Y) + H(X | Y, Z) \\ &\leq H(Z) + H(X | Y, Z). \end{aligned}$$

When $Z = 0$, the guess is correct: $X = Y$. Hence

$$H(X | Y, Z = 0) = 0.$$

When $Z = 1$ and $Y = y$, the value $X = y$ is excluded, leaving at most $|\mathcal{X}| - 1$ possibilities. Proposition 7.1 therefore gives

$$H(X | Y = y, Z = 1) \leq \log(|\mathcal{X}| - 1).$$

Averaging over Y and the two values of Z yields

$$H(X | Y, Z) \leq P_e \log(|\mathcal{X}| - 1).$$

Finally, Z is Bernoulli with parameter P_e , so $H(Z) = h_2(P_e)$. ■

Your turn. Recall the two cases that make the proof work:

$$Z = 0 : \quad X = Y, \quad H(X | Y, Z = 0) = \underline{0},$$

$$Z = 1 : \quad X \text{ has at most } \underline{|\mathcal{X}| - 1} \text{ possible values.}$$

? Correct guess: no uncertainty.
Wrong guess: at most $|\mathcal{X}| - 1$ alternatives remain.

Corollary 8.3 (*one-bit form*). Since binary entropy is at most one bit,

$$H(X | Y) \leq 1 + P(X \neq Y) \log(|\mathcal{X}| - 1).$$

Swapping X and Y gives the companion bound

$$H(Y | X) \leq 1 + P(X \neq Y) \log(|\mathcal{X}| - 1).$$

Proof. Use $h_2(P_e) \leq 1$ in Theorem 8.2. For the second inequality, apply the same theorem to (Y, X) ; the error event $\{Y \neq X\}$ and its probability are unchanged. ■

How to read the bound

The direct reading is

$$P_e \text{ small} \implies H(X | Y) \text{ small.}$$

A more useful reverse reading comes from solving the same inequality for P_e . When $|\mathcal{X}| > 2$, Corollary 8.3 becomes

$$P_e \geq \frac{H(X | Y) - 1}{\log(|\mathcal{X}| - 1)}.$$

Thus a large residual uncertainty certifies that no estimator based on Y can have a very small error probability. These are inequalities, not equalities: slack can enter through $h_2(P_e) \leq 1$ and through the bound of $\log(|\mathcal{X}| - 1)$ on the wrong-guess case.